System Analysis and design (CSE-3529)

(Mid)

Chapter-1 : Concepts of system and its environment

Chapter-2 : Information gathering

Chapter-3 : Feasibility study and Cost-Benefit Analysis

Concepts of system and its environment

- system analysis, system design, system analysis and design
- Data, information, knowledge,
- Sources of information, Types of information
- Quality/ characteristic of information
- Parts/ characteristic of system
- Information system, Types
- Relationship between different management group
- system analyst
- Role, Qualities of system analyst
- Stakeholders perspective of information system

System analysis: System analysis consists of those activities that enable a person to understand and specify what the new system should accomplish.

⇒ It describes in detail the 'what' that a system must do to satisfy the need or to solve the problem.

System analysis
- System study
- Requirement analysis [system কাজ করার জন্য যা লাগবে তা analyse করা]
- Documentation [কোনো কাজ যাতে ভবিষ্যতে কাজের ধারাবাহিকতা হারিয়ে না ফেলে তা লিখিতভাবে রাখা]

System: It is a group of elements or components which work together to accomplish a common task.

☐ System analysis and design is a study in which we learn how to analyze an existing system and create a better one.

⇒ We need it for system development

1. Creating a new one
2. Updating the existing one

System design: System design describe 'how' the system will work. It specifies in detail all the components of the solution system and how they work together to provide the solution.

⇒ It is the process of defining the architecture, components, models and data for a system to

satisfy specified requirements.

System analysis and design:

⊛ It is a step by step process for developing high quality information system.

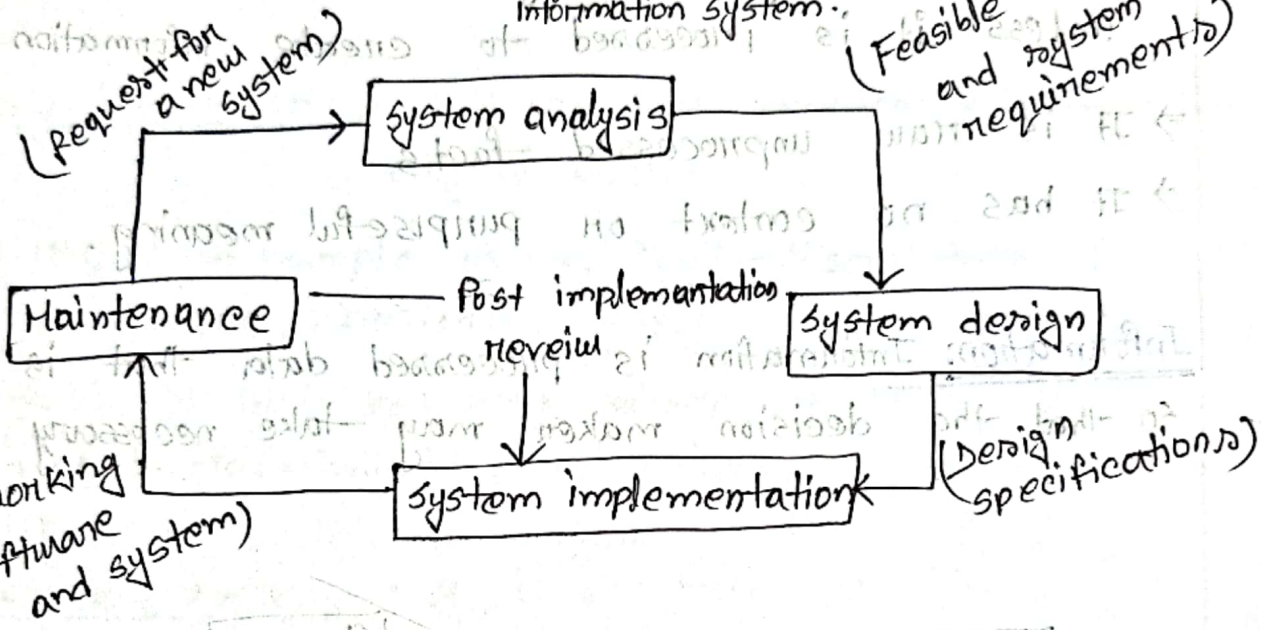


Figure: Stages in building an improved system

⇒ System development (two major components)

1. System Analysis
2. System design

⇒ After the proposed system is analyzed and designed,
- The actual implementation of the system occurs.

⇒ After implementation: ⇒ Working system is available
⇒ It requires timely maintenance

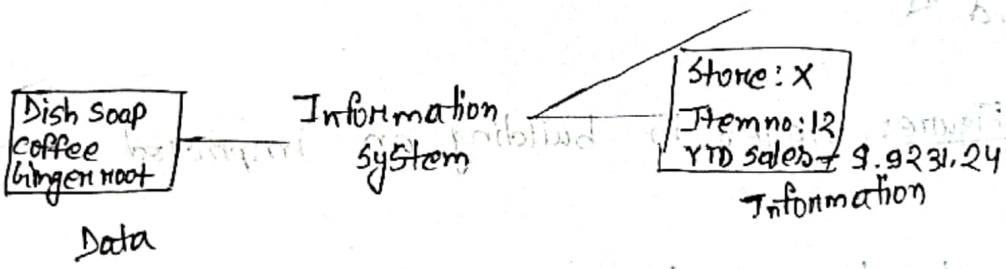
Data: Data is a collection of information gathered by observations, research or analysis.

⇒ Data is generally used by machines and is useless unless it is processed to create information.

⇒ It is raw unprocessed facts

⇒ It has no context or purposeful meaning.

Information: Information is processed data that is useful so that the decision maker may take necessary action.



Knowledge: Knowledge is awareness or understanding of something such as facts, information, skills which is acquired through experience or education by learning.

⇒ Data and information that is further refined based on the facts, truths, beliefs, experiences and expertise of the recipient.

Sources of information:

1. Primary information
2. Secondary information
3. Internal information
4. External information

Primary information: ⇒ It is one that provides data from an original source.

Example:
This maybe as simple as an invoice sent to a business or a cheque received.

⇒ It is defined as being where a piece of information appears for the 1st time.

Secondary information: ⇒ It is one that provides information from a source other than the original.

⇒ Secondary sources are processed primary sources, second-hand versions.

Example: A bank statement that shows details of cheques paid in and out.

Internal information: All organizations generate a substantial amount of information relating to their operation. The information maybe available from a number of sources within the organisation. This internal information is vital to the successful management.

Example: Marketing and sales information on performance, revenues, financial information on profits, cost etc.

External information: An external source of information is concerned with what is happening beyond the boundaries of the organisation.

Example: Telephone directories, professional publications etc.

Types of information:

Strategic information:

(long range) — लम्बी अवधि के लिए planing

⇒ It is used at the very top level of management.
⇒ It is broad based and will use a mixture of information gathered from both internal and external sources.
⇒ Timescale maybe from 1-5 years or even longer depending on the project.

Tactical information:

(Short range)

⇒ It is used by middle management when managing or planning Projects
⇒ The next level down is the tactical level, and tactical planning and decision making takes place within the guidelines.
⇒ Timescale is between 6 month and 5 years (depending on Strategic project)

⇒ Strategy refers to the 'what' and 'why' a company plans to do in the future. And tactical refers to 'how' it plans to implement it.

operational information:

(very short)

⇒ The lowest level is operational and operational planning takes place based on the tactical plans.
⇒ Timescale is usually very short (daily or at most a week or month)
⇒ Needed for day to day operations of the organization
⇒ Results of operational work will usually be passed upwards to let the tactical planners to evaluate their plans.

System Support	Management level	Type of information	Information level
DSS	upper	unstructured	Strategic Long range planning
MIS	Middle	Moderately structured	Tactical short range improvement
DPS	lower	highly structured	operational operational day to day policies

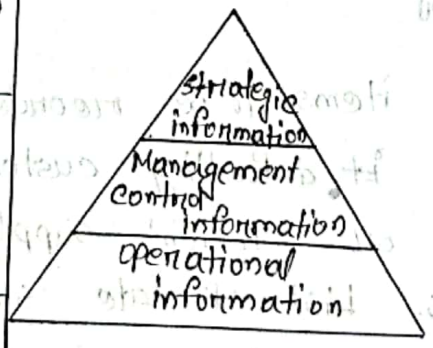


Figure: Management hierarchy and information needs

Quality / Characteristics of information:

1. **Availability / accessibility:** Information should be easy to obtain on access. Example: Telephone directory in internet.

2. **Accuracy:** Information needs to be accurate enough for the use to which it is going to be put. The degree of accuracy depends upon the circumstances.

3. **Reliability:** It deals with the truth of information or the objectivity with which it is presented.

4. **Relevance / appropriateness:** Information should be relevant to the purpose for which it is required.

5. **Completeness:** Information should contain all the details required by the user. Otherwise it may not be useful as the basis for making a decision.

6. **Level of detail / conciseness:** Information should be in a form that is short enough to allow for its examination and use.

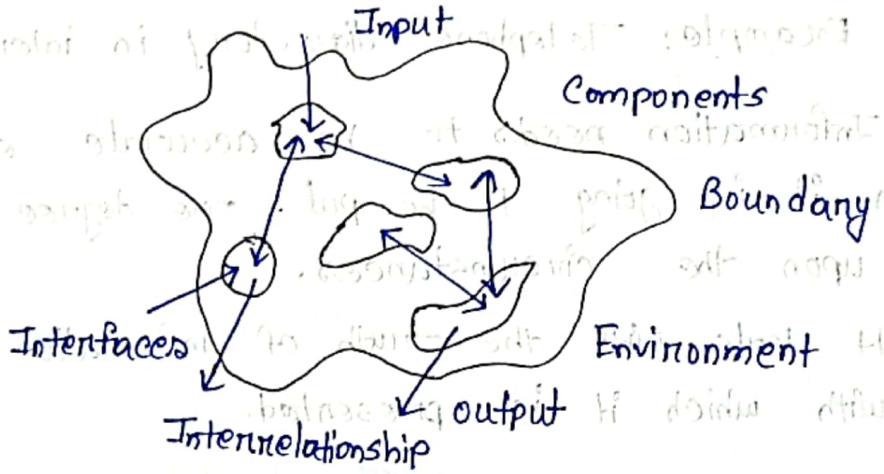
7. **Presentation:** Use attractive format and graphical charts.

8. **up-to-date:** Include all data up to present time.

9. **Timing:** Information must be on time when it is required. Information received too late will be irrelevant.

Parts of system / Characteristic of system

A system has 9 characteristics



① Component: It is either an irreducible part or an aggregate of parts also called a subsystem.

② Interrelated components: Components are interrelated. The function of one is somehow tied to the functions of the other systems.

④ Purpose: Overall goal ~~for~~ or function of a system.

⑤ Environment: Everything outside the system's boundary that influences the system.

⑥ Interface: The system needs its environment are called interfaces. An interface also occurs between subsystems.

⑦ Constraints: The limits (capacity, speed, capabilities) to what it can do.

⑧ Output: output transferred the process information.

⑨ Input: Input captures or collect raw data from within the organization.

③ Boundary:

Important system concepts:

Decomposition: It is a technique that allows the system analyst
- ⇒ Break a system into small, manageable, and understandable subsystems
- ⇒ Focus attention on one area at a time.

Modularity: ⇒ It is a direct result of decomposition.
- ⇒ It refers to dividing a system into modules.
- ⇒ Modules can represent a system ~~easily~~ simply, making it easier to understand.

Coupling: ⇒ It means subsystems are dependent on each other.
- ⇒ If one subsystem fails and other highly dependent on it the other will fails themselves.

Cohesion: It is the extent to which a subsystem performs a single function.

Information system: Information system defined as a set of devices, Procedures and operating systems designed around user based criteria to produce information and communicate it to the users for planning, control and performance.
- ⇒ It is a formal, organizational system designed to collect, process, store and distribute information.

Why information systems is necessary?

~~1. Foundation of~~

4 reasons why IS makes a difference to the success of a business.

1. Foundation of doing business:

⇒ It can help a business become a high-quality, low-cost producer.

⇒ It is vital to the development of new products.

2. Capital Management:

⇒ The success of your business in the future may well depend on how you make it investment decisions.

3. Productivity: IT has increased productivity in the economy by about 1% in the last decade.

IT is a major source of labor and capital efficiency.

4. Strategic opportunity and Advantage:

⇒ Create competitive advantage

⇒ New business models

⇒ Create new services.

Types of information system:

1. Transaction processing systems (TPS)
2. Management information system (MIS)
3. Decision support system (DSS)
4. Executive support system (ESS)
5. Enterprise systems
6. Supply chain management system (SCM)
7. Customer relationship management (CRM) system

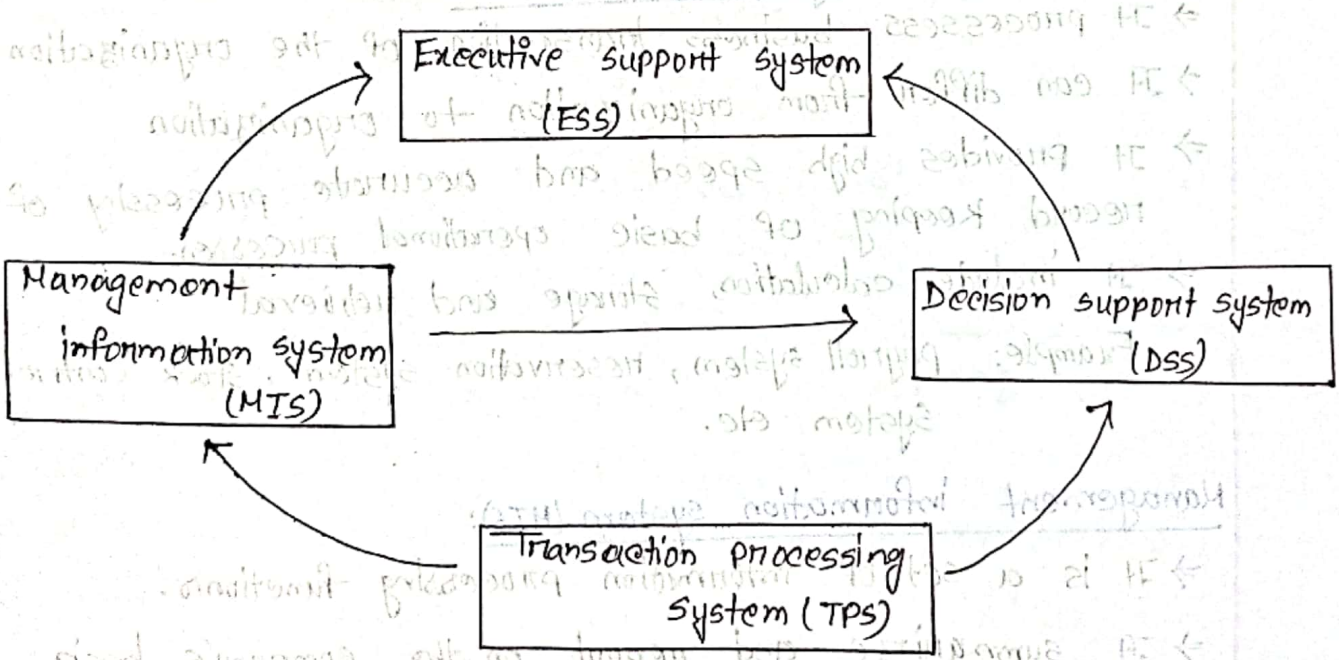
} System for different Management Group

} System for linking the enterprise

8. Expert system
9. knowledge management system

} Other information system

Relationship between different management group:



Transaction processing system: (TPS)

⇒ It processess business transaction of the organization.
⇒ It can differ from organization to organization
⇒ It provides high speed and accurate processing of record keeping of basic operational processes.
⇒ It include calculation, storage and retrieval
Example: payroll system, reservation system, stock control system etc.

Management information system (MIS):

⇒ It is a set of information processing functions.
⇒ It summarize and report on the company's basic operations using data supplied by TPS.
⇒ These system assist lower management in problem solving and making decisions.
Example: Budgeting system, sales management system etc.

Decision support system (DSS): ⇒ It support semi structured and unstructured decision making.

⇒ use of analytical and simulation models. Artificial intelligence tools maybe used

⇒ It is a computer program application used to improve a company's decision making capabilities.

⇒ It analyzes loarge amount of data.

Example: Financial planning systems, Logistics system etc.

Executive support system (ESS): ⇒ It allows users to transform enterprise data into quickly accessible.

⇒ ESS rely on the information generated by TPS, MIS and DSS.

⇒ It present graph and data from many sources.

Example: External database,

Enterprise system: ⇒ It is known as enterprise resource planning (ERP)

⇒ It helps to improve coordination and decision making.

Supply chain management system (SCM): ⇒ Information systems that automate the flow of information between a firm and its suppliers.

⇒ It is widely used by manufacturing and distribution firms to connect to suppliers.

⇒ The ultimate goal is to get the right amount of the products.

Example: Retail customer, Manufacturer etc.

The farms
↓
Food processing plants → Freight companies → Distribution centers → Freight companies → Restaurants

CRM: Information systems that track all the ways in which a company interacts with its customers and analyze those interactions to optimize revenue, customer satisfaction.

Expert system: It is a computer system that emulates the decision making ability of a human part.

⇒ This systems are designed to solve complex problems.

⇒ Expert system is divided into two subsystems:

1. Interface engine, 2. knowledge base

Example: Artificial intelligence system that emulates an auto ~~machine~~ mechanic's in automobile problem

knowledge management system (KMS): ⇒ It enable organizations to better manage processes for capturing and applying knowledge and expertise.

⇒ These system collect all relevant knowledge.

System Analyst: A system analyst investigates, analyzes, designs, develops and maintains a company's information systems.

Role of system analyst / Responsibilities:

⇒ It helps translate business requirements into IT project

⇒ An analyst might help document business profile, design information system, ~~we~~ websites etc.

⇒ A systems analyst must plans projects, develops

Schedules and estimates ~~code~~ costs:

⇒ It differs from organization to organization,

Some common roles:

1. Investigator, 2. Monitor, 3. Architect, 4. Motivator,
5. problem solving, 6. psychologist.

Qualities of system analyst:

Technical knowledge: State of the art knowledge is extremely important.

⇒ Analyst also maintain their skills by attending training courses, both on side and online.

Communication skills: The analyst must work with people outside the company, A system analyst needs strong oral and written communication skills.

Business skills: A system analyst works closely with managers, supervisors. He/she must understand business operation, communicate clearly, and translate business needs into requirement that can be understood by programmers and system developers. A successful analyst is business oriented

Critical thinking skill: ⇒ It include the ability to compare, classify, evaluate, analyzed and apply logic.

⇒ Critical thinking skill are valuable in IT industry.

⇒ They often use a what if approach, and they have the ability to evaluate their own thinking and reasoning.

Education: For higher level positions, many companies require an advanced degree.

companies typically require systems analysts to have a college degree in information system.

Stakeholders perspectives on an information system:

1. System owners — Responsible for funding the project developing, operating and maintaining the information system.
2. System users — • Internal system users
   - ⇒ clerical and service workers
   - ⇒ Technical and professional staff
   - ⇒ Supervisors, executive manager

   • External system users
   - ⇒ customers, ⇒ suppliers, ⇒ partners, ⇒ employees

   ⊞ Remote users: who are not physically located but who still require access to information system.

   ⊞ Mobile users: whose location is constantly changing but who requires access to information systems from any location.
3. System designer
4. system builders
5. systems analyst
6. External service provider (ESP)
7. Project manager.